

Jonas Kemnitzer

Waldstraße 1, 95183 Feilitzsch, Germany
jonaskemnitzer@pm.me | +49 172 692 7004

EDUCATION

UAS HOF

M.Sc. IN COMPUTER SCIENCE
August 2023 | Hof, Germany

KEIMYUNG UNIVERSITY

EXCHANGE SEMESTER
September 2019 | Daegu, S. Korea

UAS HOF

B.Sc. IN COMPUTER SCIENCE
March 2021 | Hof, Germany

COURSEWORK

GRADUATE

Modern Software Architecture
Information Systems Security
Data Mining & Machine Learning,
IT & Data Protection Law
Development of Business Models

UNDERGRADUATE

Software Engineering
Algorithms & Data-Structures
Math & Statistics for CS
Applied Machine Learning
RESTful Web Services
Robotics
IT-Security
Geographic Information Systems
Virtual Reality Applications
Java Programming
Korean Language

SKILLS

TECHNICAL SKILLS

• Python • Streamlit • OOP
• Git • Software Engineering
• SQL-DB's • Vector-DB's
• Machine Learning
• PyTorch • VSCode • $\LaTeX$

SOFT SKILLS

• Agile Workflows
• Trilingual (German, English, Korean)
• Cross-Team Working
• Analytical Thinking & Adaptability

CLUBS

UNIVERSITY RELATED

• Member of Students Union
• Honorary intercultural teacher

HOBBY RELATED

• Trainer at TC Hof

WORK EXPERIENCE

INSTITUTE FOR INFORMATION SYSTEMS | RESEARCH ASSOCIATE

August 2023 – Now | Hof, Germany

- Conducted research for the Intelligent and Learning Systems research group, focusing on applications of artificial intelligence, including machine learning and deep learning, in multi-modal environments.
- Collaborated with a multidisciplinary team to design and execute experiments, contributing to published research papers in AI and robotics conferences.
- Lectures for undergraduate and graduate students, providing guidance on AI and machine learning projects.

IBM | AI PROTOTYPE ENGINEER

April 2020 – August 2020 | Munich, Germany

- Implemented supervised and unsupervised ML algorithms for analysis, visualization, and forecasting purposes.
- Improved AI life-cycles using IBM Cloud and Python/Spark by around 10%.
- Collaborated with solution architects improving customer-oriented demos.
- Restructured message BUS between a *Pepper* robot and IBM Watson Assistant creating engaging dialogue options.

PROJECTS

FULL STACK PROJECT | DEVELOPMENT

Analysis and implementation of a full-stack timetabling software in a team of 10 people using technologies like MySQL and *Spring Boot* for backend functionality as well as *Angular* for creating a modern UI.

- Evaluated pros and cons between 3 different stack solutions.
- Implemented frontend and backend components.
- Held code reviews with team members from other departments.

EXPLAINABLE AI (LIME) | RESEARCH

Theoretical analysis as well as the implementation of the *Local Interpretable Model-agnostic Explanations (LIME)* algorithm making use of technologies like Python and Google Collab Notebook.

- Presented and discussed results in front of around 20 people.
- Optimized input parameters to reduce data complexity.

SENTIMENTAL ANALYSIS USING NLP | RESEARCH

Development of a proof of concept artificial intelligence demo as part of the sentimental analysis of an online store using Natural Language Processing with Python, TensorFlow, Keras and the Google Collab Notebook.

- Tested different frameworks to measure performance differences.
- Reduced code base to improve overall runtime performance.

PUBLICATIONS

- [1] J. Kemnitzer and C. Groth. Enhancing fitness visualization: application and efficacy of realistic inpainting techniques using diffusion models. In *International Conference on AI-Generated Content (AIGC 2024)*, volume 13649, page 1364907. International Society for Optics and Photonics, SPIE, 2025.